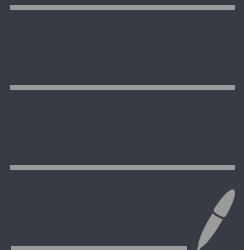


Scaling Law Notebook

replacing the back
of the cover



Self-Attention

$$Q \in \mathbb{R}^{T \times D}$$

$$K \in \mathbb{R}^{T \times D}$$

T = sequence length (query) i.e. number of tokens

embedding dim $\neq q, k, v$ dim. e.g. 64

D = feature dim. of each token's representation e.g. 512

visualisation

length of longest sentence in
training data set

$$Q = \begin{bmatrix} q_1 & \dots & q_{1D} \\ q_2 & \dots & q_{2D} \\ \vdots & \ddots & \vdots \\ q_T & \dots & q_{TD} \end{bmatrix}$$

T rows, T number of tokens

D columns, embedding dimensions, each row has D numbers

$$QK^T \rightarrow \text{transposed} \in \mathbb{R}^{T \times T}$$

$$Q = \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix} \quad K^T = [k_1, k_2, k_3]$$

$$QK^T = \begin{bmatrix} q_1 k_1 & q_1 k_2 & q_1 k_3 \\ q_2 k_1 & q_2 k_2 & q_2 k_3 \\ q_3 k_1 & q_3 k_2 & q_3 k_3 \end{bmatrix}$$

(i,j) th
element

$$(QK^T)_{ij} = q_i k_j$$

$$\text{e.g. } (2,3) = q_2 k_3$$

$$\star X \cdot W^Q = Q \in \mathbb{R}^{T \times D}$$

$$X \cdot W^K = K \in \mathbb{R}^{T \times D}$$

$\in$ = belongs to / is an element of

$$\mathbb{R}^{X \times Y} = XY \text{ real numbers}$$

$T = S$
 ↓ sequence length (key value)

e.g. $T=6$

$$Q \in \mathbb{R}^{6 \times D}$$

$$K \in \mathbb{R}^{6 \times D}$$

$$QK^T \in \mathbb{R}^{6 \times 6}$$

each query gets a score against these blocks.

assume D
 is 7

$$\begin{bmatrix} q_1 \\ q_2 \\ q_3 \\ q_4 \\ q_5 \\ q_6 \end{bmatrix}$$

$T=6$ in self-attention

$$\cdot [k_1, k_2, k_3, k_4, k_5, k_6]$$

$$= \begin{bmatrix} q_1 k_1 & q_1 k_2 & q_1 k_3 & q_1 k_4 & q_1 k_5 & q_1 k_6 \\ - & - & - & - & - & - \\ - & - & - & - & - & - \\ - & - & - & - & - & - \\ - & - & - & - & - & - \end{bmatrix}$$

$$6 \times 6 = 36 \text{ scores}$$

★ score meaning

$q_1 k_1$ the relevance of the first token to the first token when decoding
 $q_1 k_2$ the relevance of the second token to the first token when decoding

Attention equation

$$\text{Attention}(Q, K, V) = \text{softmax}(QK^T)V$$

↳ stays the same shape $[T, T]$ afterwards

$$V \in \mathbb{R}^{T \times D}$$

$$[T, T] [T, D]$$

$$[T, D]$$

same shape as original sequence representation

$$\text{Attention}(Q, K, V) \in \mathbb{R}^{T \times D}$$

$$\text{softmax} : \text{softmax}(z_i) = \frac{e^{z_i}}{\sum_j e^{z_j}}$$

e.g. logits $\hat{=}$ raw output from model

0.1, -1.0, 2.0

$$\begin{array}{ccc} \frac{e^{0.1}}{(e^{0.1} + e^{-1} + e^2)} & \frac{e^{-1}}{(e^{0.1} + e^{-1} + e^2)} & \frac{e^2}{(e^{0.1} + e^{-1} + e^2)} \\ \approx 0.12 & \approx 0.04 & \approx 0.83 \end{array}$$

sum = 1

$$\text{Temperature softmax} : \text{softmax}(z_i) = \frac{e^{z_i/T}}{\sum_j e^{z_j/T}}$$

e.g. $T = 0.8$

small $\leftarrow T \rightarrow$ large
 $\downarrow$ $\downarrow$
 deterministic creative

$$\frac{e^{0.1/0.8}}{e^{0.1/0.8} + e^{-1/0.8} + e^{2/0.8}} + \frac{e^{-1/0.8}}{e^{0.1/0.8} + e^{-1/0.8} + e^{2/0.8}} + \frac{e^{2/0.8}}{e^{0.1/0.8} + e^{-1/0.8} + e^{2/0.8}}$$

Q is the sentence we're looking at.

K is kinda like a memory address, it doesn't contain the information.

V is the actual information.

Attention

$$Q[T, D] \times K^T[D, i] \rightarrow [T, i] \xrightarrow{\text{softmax}} [T, i] \times V[i, D] \rightarrow [T, D]$$

$\downarrow$

if $T = 100000$

then $T, i = 10^{10}$ which is very enormous

Flash attention wants to calculate $\text{softmax}(QK^T)V$ without storing enormous attention weights.

MoE models

✗ single dense MLP block per layer → ✓ sets of ind. MLPs that can be dynamically routed between

§ A normal dense model with E MLP blocks per layer

each token activates k of these experts ie MLP blocks, typically $k < E$

Ratio $\frac{E}{k}$ sparsity usually $8 < 2 < 64$

e.g. DeepSeek v3 $\frac{256}{8} = 32$

ie 256 experts/MLP blocks per layer, each token activates 8 of them

§ Parameter count = $E * \text{size of each expert}$

§ Not all experts are used for each token

Block Remat is materialize this value again by recomputing it
↳ rematerialization

↳ only save the input of the layer, so you can rerun it again during backprop.

memory is expensive, compute is cheaper

$$\begin{aligned} 7 \text{ check point per layer} &= \frac{84}{20} \quad \text{vs} \quad 20 \text{ check points per layer} = 84 \text{TB} \\ &= 4.2 \text{TB} \end{aligned}$$

Attention FLOPs : $12BT^2NH$

matmul FLOPs : $18BTD^2 + 24BTDNH$

matmul FLOPs + Attention FLOPs - MLP training FLOPs

$$= 18BTD^2 + 24BTDNH + 12BT^2NH - 18(BTD)^2$$

$$= 24BTDNH + 12BT^2NH$$

$$= \text{Attention training FLOPs}$$

B batch size
T Q Sequence length
S K/V sequence length
N no. query heads
K no. KV heads
H head dim.
Gr no. query heads per kv head